EX.NO: 03
DATE: 30/09/26

Determination Voltage In circuit using Nodal Analysis

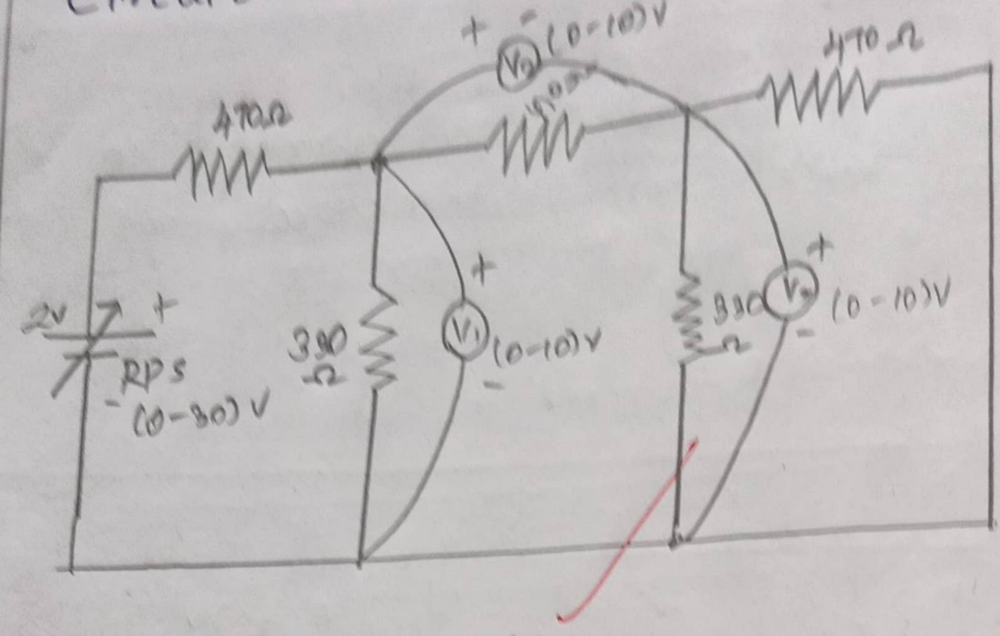
AIM:-

To determine the voltage in the circuit using nodal analysis both theoretically and practically for a given DC circuit.

Apparatus Required :-

S.NO	Apparatus	specification	Quantity
1	RPS	(0-30)V	1
2	Multimeter	-	3
3	Resistor	470Ω, 330Ω	3, 2
4	Bread board	-	1

Circuit Diagram :-



Observation :-

~~S.No~~ / Parameter RPS(volts)	Theoretical value			practical value		
	V_A	V_B	$V_A - V_B$	V_A	V_B	$V_A - V_B$
2	0.65	0.2	0.45	0.69	0.2	0.48
4	1.27	0.37	0.9	1.33	0.39	0.9
6	1.91	0.55	1.35	2.0	0.68	1.41
8	2.6	0.8	1.8	2.64	0.77	1.86

Calculation :-

$$\frac{V_A - 2}{470} + \frac{V_A}{330} + \frac{V_A - V_B}{470} = 0 \longrightarrow ①$$

$$\frac{V_B - V_A}{470} + \frac{V_B}{330} + \frac{V_B}{470} = 0 \longrightarrow ②$$

By solving ① & ②,

$V_A = 0.65$

$V_B = 0.20$

$V_A - V_B = 0.45$

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RESULT :-

i) Voltage (V_1) = 0.65
ii) Voltage (V_2) = 0.45
iii) Voltage (V_3) = 0.2

} for (RPS = 2V)